HENRY DEUTSCH

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Portfolio: **henry-deutsch.com**

WORK EXPERIENCE

NewForm AI | *Software Engineer I — Full Time* Aug 2025 - Mar 2026

Early engineer at a B2B ad-analytics startup serving 50+ consumer brands; reported to the CTO and owned 6 initiatives

- Built LLM agent harness for WillBot, enabling natural-language analysis and decision support across $100M+ in paid media spend. Automated regression testing across single- and multi-client queries to improve agent reliability.
- Led major refactor making the platform ad-network-agnostic, unlocking $50K+/mo in TikTok spend. Refactored 31K+ LOC across 140+ tRPC endpoints and rebuilt the Bayesian inference engine around platform-agnostic objects
- Built and migrated 100% of internal account managers and external clients to a streamlined messaging and reporting system

KnoWhiz | *Software Engineering Intern* May 2024 - Sep 2024

Developed full-stack features for LLM-based AI learning platform that serves 4.3K+ monthly active users

- Rearchitected full-stack Quizlet import feature to cut large import failures from ~50% to <4% on 4.3K MAUs, saving ~200 user hours /mo. Used WebSocket-based progress updates, Java Spring Boot batching and two-phase commit patterns
- Led SEO improvements and created a gateway Explore page, increasing monthly organic traffic by ~10%
- Mentored developer while owning the import rearchitecture, reviewing 40+ of their PRs through to merge

HF Engineering | *Software Engineering Intern* May 2024 - Sep 2024

Led development of internal onboarding platform enabling 18,000 volunteers to collaborate on open source projects

- Used AWS Amplify for CI/CD; coordinated Azure DevOps releases with feature flags and rollbacks across 8 teams.
- Created full-stack dashboard for project management with React (Next.js) and JavaScript, with user authentication

Johns Hopkins University | *Software Development Intern* May 2023 - Aug 2023

Built full-stack disaster relief analytics platform for the National Institute of Standards and Technology (NIST)

- Integrated Envoy with Istio and Prometheus to add tracing, telemetry, and observability across REST APIs and microservices
- Led database redesign and PostgreSQL schema optimization to support new GPS analysis features
- Designed Go microservice ingesting disaster-response sensor data at 1k msg/sec with 250ms latency for real-time dashboards

EDUCATION

Johns Hopkins University | Bachelor of Science in Computer Science | Baltimore, MD Aug 2021 - May 2025

Graduated with Departmental Honors in Computer Science

Courses: Full-Stack JavaScript, Artificial Intelligence, Machine Learning, Practical Generative AI, Computer Graphics, Intermediate Programming (C++), Computer System Fundamentals, Data Structures, Algorithms, Calculus III

Awards: Placed 3rd of 43 teams, won $250 prize at HopHacks Hackathon for reimagining grocery checkout with computer vision

SOFTWARE PROJECTS

Chief of NYC Fire Dept Website | Paid Contract — Full Stack Developer Sep 2021 - Sep 2022

- Remade website, increasing book sales $16K YoY (tripling sales) for client; site traffic went from <100 to >3.6k visits/mo
- Designed custom Java-based PDF parser to extract structured content from 200+ pages and generate Next.js layouts
- Developed learning platform with React and JavaScript with quiz functionality, content management, and progress tracking

C++ Ray Tracing Engine

- Architected ray tracing engine with C++, OpenGL, and multithreading — achieved 2.5x speedup over single-thread approach
- Engineered 3D rendering pipeline with GLSL shaders and parallel computational geometry intersection algorithms
- Developed Phong illumination with quaternion camera controls, bilinear texture mapping, and ray-traced shadow systems

N-Body Orbit Simulations | Research at Johns Hopkins Aug 2023 - May 2024

- Built convolutional autoencoder with PyTorch and NumPy. Achieved 200:1 compression with .02 MSE loss on orbit data
- Classified orbit paths by their latent space in the NN with K-means, Hierarchical, and Agglomerative clustering in Python
- Deployed ML infrastructure on AWS EC2 and GCP Compute Engine for parallel processing of 10k+ orbit simulations

TECHNICAL SKILLS

Full-Stack Web Dev: Next.js, React, Node.js, Express.js, Prisma, Redis, tRPC, MongoDB (MERN stack), SQL, Tailwind, Vite
Languages: TypeScript, JavaScript, Python, Java, C++, Go, SQL, HTML, CSS
Other: Azure DevOps, Prometheus, Linux, Kubernetes, Docker, AWS (EC2, Amplify), GCP, CI/CD, Git, GitHub, Agile/Scrum